Computational Linear Algebra

n ppl told to choose H or T for order of 8 coin tosses
After each round, ppl who choose wrong toss get eliminated

$n \rightarrow$ ppl
$x \rightarrow$ rounds | How many rounds such that no one remains?

$P(H/T) = 1/2$

$\frac{n}{2^x} < 1 \Rightarrow n < 2^x$

$\log n < x \log 2$

$\boxed{x > \log_2 n}$

given $A^{16} = A * A * \ldots * A$ (15 multiplications)
Reduce the number of multiplications.

$x = \log_2 16 = 4 \Rightarrow$

$A^{16} = A^8 * A^8$
$A^8 = A^4 * A^4$
$A^4 = A^2 * A^2$
$A^2 = A * A$

} total 4 multiplications used.

⊛ **Rank of Matrix**: order of largest square sub matrix whose determinant $\neq 0$

→ SINGULAR $\Rightarrow$ det = 0
→ NON-SINGULAR $\Rightarrow$ det $\neq 0$

- rank(In) = n
- rank(null matrix) = 0
- $r(A) = r(A^T)$
- if $A \neq$ null matrix $r(A) \geq 1$

⊛ **Eigenvalue and Eigenvectors**

→ $AX = \lambda X$ [A is any square matrix. We can find a column matrix X and constant λ]

only scaling no rotation.

$|A - \lambda I| = 0$ [Characteristic Eqn]

Eigenvalues $\Rightarrow$ values of λ, after solving $|A - \lambda I| = 0$

Eigenvectors $\Rightarrow X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ for each corr. eigenvalue λ

⊛ for 2×2 matrix $\Rightarrow \lambda^2 - tr(A)\lambda + |A| = 0$

⊛ for 3×3 matrix $\Rightarrow \lambda^3 - tr(A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$

⊛ sum(eigenvalues) = trace(A)
⊛ ~~sum~~ prd(eigenvalues) = det(A)
⊛ $A \rightarrow \lambda$, $A^{-1} \rightarrow 1/\lambda$
⊛ if A is orthogonal i.e $A^T A = A A^T = I$ i.e $A^T = A^{-1}$, then $\lambda, \frac{1}{\lambda}$ are eigenvalues.

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$\boxed{AX = \lambda X}$ $X \rightarrow$ eigenvector of A corr. to eigenvalue λ

$$\underset{A}{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} \underset{P}{\begin{bmatrix} e & f \\ g & h \end{bmatrix}}$$

$= A[X, Y]$

$= [AX, AY]$

$= [\lambda_1 X, \lambda_2 Y]$

$= \begin{bmatrix} \lambda_1 e & \lambda_2 f \\ \lambda_1 g & \lambda_2 h \end{bmatrix}$

$= \begin{bmatrix} e & f \\ g & h \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

$$A \underbrace{[X, Y]}_{P} = \underbrace{[X, Y]}_{P} \underbrace{\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}}_{D}$$

$$AP = PD$$

$$\boxed{\begin{aligned} A &= PDP^{-1} \\ A^n &= PD^nP^{-1} \end{aligned}}$$

DIAGONALISATION OF A MATRIX

[EIGEN/SPECTRAL DECOMPOSITION]

I)

⊛ Linear Transformations

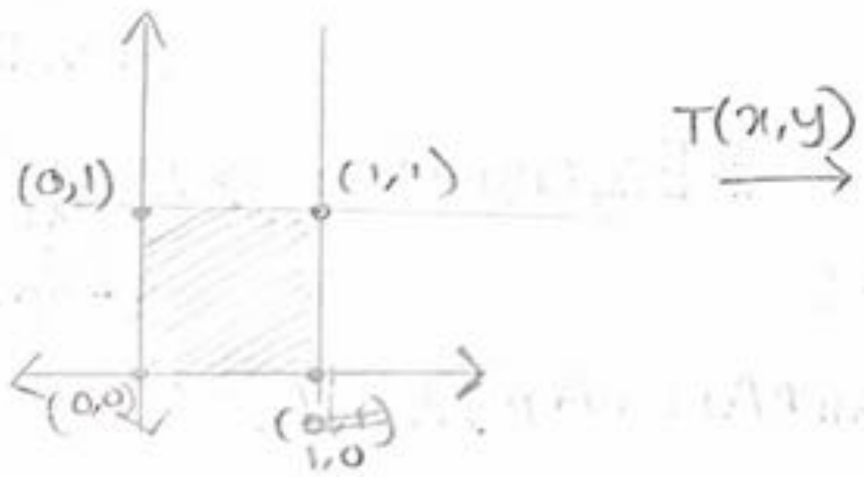
$T: V \rightarrow W$ is said to be a L·T if T proves:

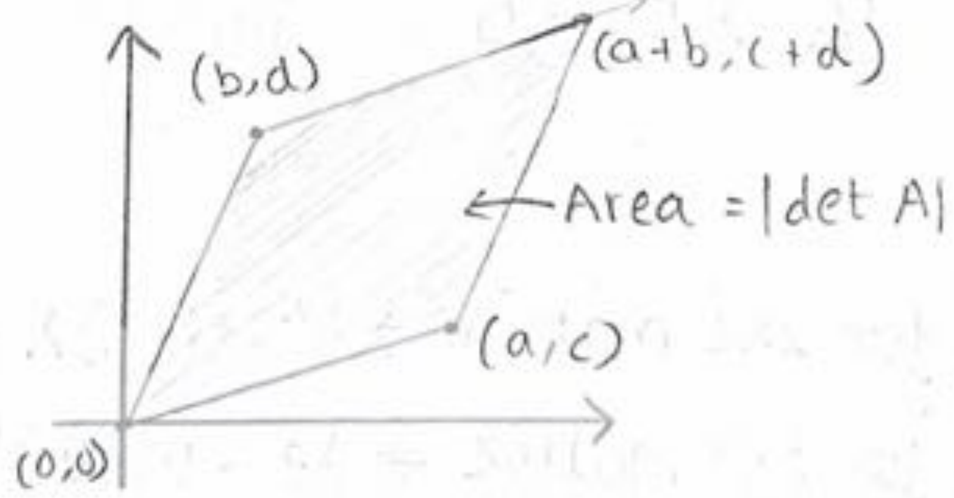
(i) addition $T(X+Y) = T(X) + T(Y) \quad \forall x, y \in V$

(ii) scalar multiplication $T(\alpha X) = \alpha T(X), \forall x \in V$

i.e $\boxed{T(\alpha X + \beta Y) = \alpha T(X) + \beta T(Y)}$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax+by \\ cx+dy \end{bmatrix} \quad | \quad T(x,y) = (ax+by, cx+dy)$$





m(T) matrix $\longleftrightarrow$ L·T Linear Transf. T

$A^{-1} \longleftrightarrow T^{-1}$

$$\boxed{A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}}$$

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g). $T(x,y) \Rightarrow$ given

$\therefore T^n(x,y) \Rightarrow$ to find

① find eigenvectors for A i.e $\bar{X}$

② find D matrix

③ $A = PDP^{-1}$

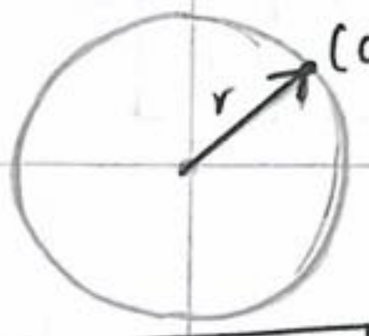
④ $\therefore A^n = PD^nP^{-1}$

⑤ $T^n(x,y) = [PD^nP^{-1}] \begin{bmatrix} x \\ y \end{bmatrix}$

II) Rotations in 2-D

Rotation in complex plane can be thought of as multiplying by 'i'

$(x,y) \equiv x+yi$



$z = re^{i\theta}$ Unit Circle

$e^{i\theta} = \cos\theta + i\sin\theta$

$e^{i\pi/2} = 0 + i$

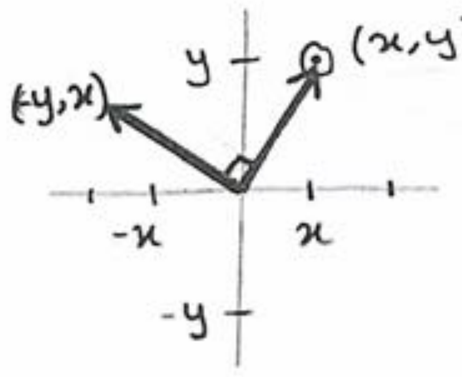
$e^{i\pi} = -1 + 0$

$\boxed{e^{i\pi} + 1 = 0}$

Multiplication by $e^{i\theta}$ rotates a vector by θ

eg: $(x+yi)i = xi - y$

$\equiv (-y, x)$



$R_{\pi/2} = i = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$R_{-\pi/2} = i^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$

☆ ☆ $a+bi = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ $\therefore e^{i\theta} = \cos\theta + i\sin\theta$

Matrix which rotates a vector (x,y) by an angle θ in anticlockwise sense is given by

$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ (a·c·w)

$R_{-\theta} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ (c·w)

⊛ $R_\theta \cdot R_\theta^T = I = R_\theta^T \cdot R_\theta$

$\therefore R_\theta^T = (R_\theta)^{-1}$ ← challenging to compute

easy to compute → $R_\theta = (R_{-\theta})^{-1}$

III) Standard Linear Transformations

① Identity

$$T(x,y) = (x, y) \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

② Scaling (about x-axis)

$$T(x,y) = (ax, y) \equiv \begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

③ Scaling (about y-axis)

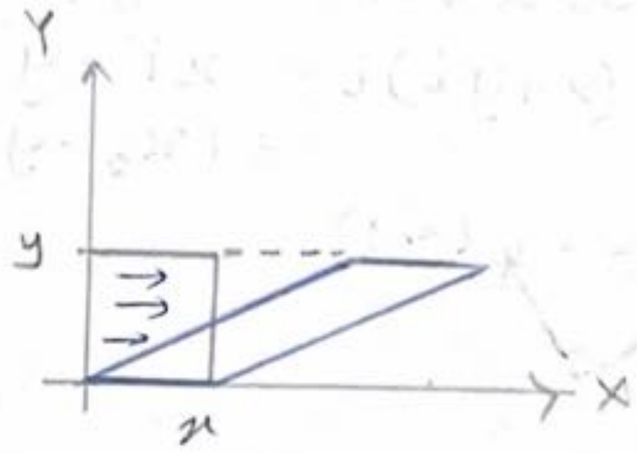
$$T(x,y) = (x, by) \equiv \begin{bmatrix} 1 & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Combining ② and ③:

$$T(x,y) = (ax, by) \equiv \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

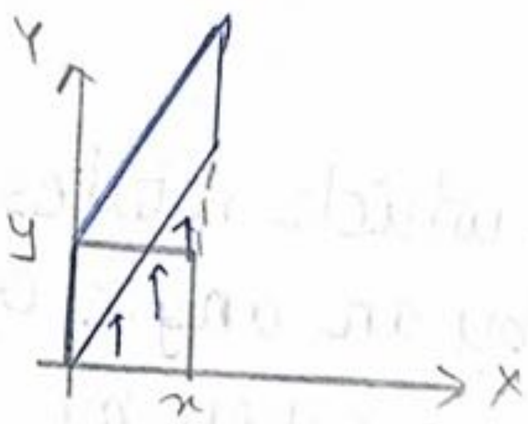
④ Shear (about x-axis) (y- fixed)

$$T(x,y) = (x + Ky, y) \equiv \begin{bmatrix} 1 & K \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



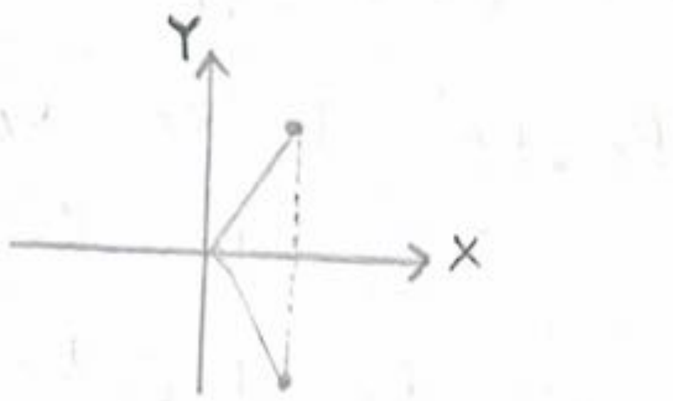
⑤ Shear (about y-axis) (x-fixed)

$$T(x,y) = (x, y + Kx) \equiv \begin{bmatrix} 1 & 0 \\ K & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



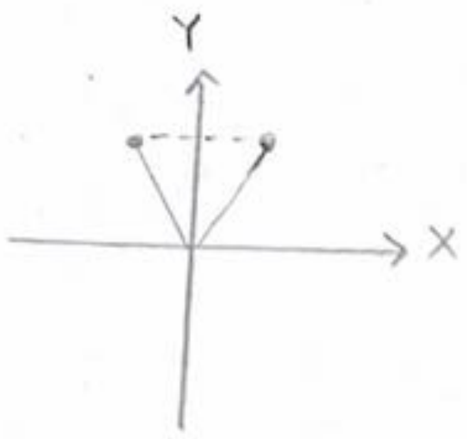
⑥ Reflection ~~about~~ w.r.t x-axis

$$T(x,y) = (+x, -y) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



⑦ Reflection w.r.t y-axis

$$T(x,y) = (-x, y) \equiv \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

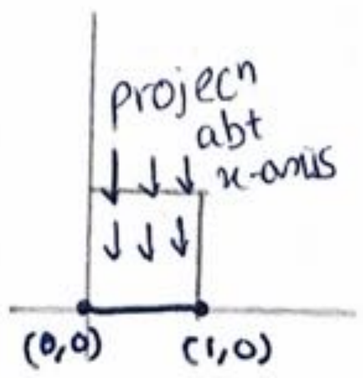


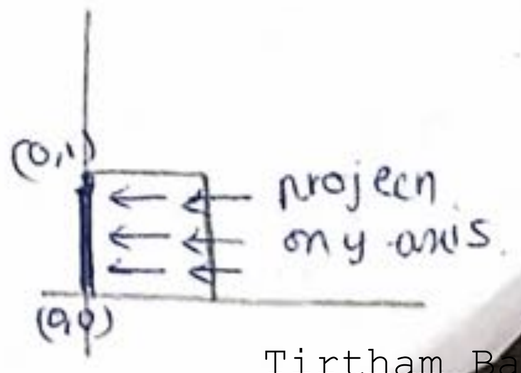
⑧ Projection on x-axis

$$T(x,y) = (x, 0) \equiv \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

⑨ Projection on y-axis

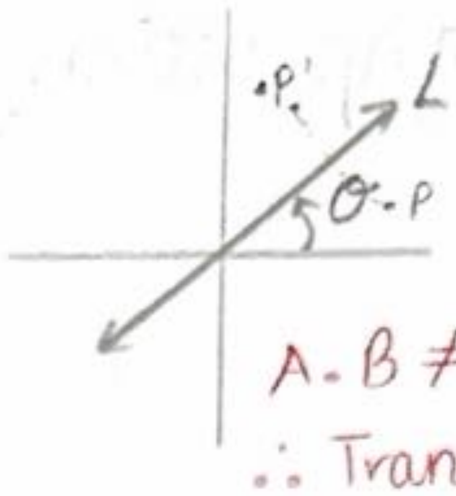
$$T(x,y) = (0, y) \equiv \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$





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(10). Reflection about line L



$A \cdot B \neq B \cdot A$

∴ Transformation Matrix will come first

∴ order should be maintained.

∵ ① rotate L clockwise by angle θ to converge/overlay with x-axis $(R_{-\theta})$

② reflect about x-axis $\left(\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\right)$

③ rotate L by angle θ (a.c.w) (R_θ)

Step 1 : $R_{-\theta}$

$$T(x,y) \xrightarrow{R_{-\theta}} \begin{bmatrix} \cos\theta & +\sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

Step 2: Reflect abt x-axis

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & +\sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$$

Step 3 : R_θ

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix} = \begin{bmatrix} \cos^2\theta - \sin^2\theta & 2\cos\theta\sin\theta \\ 2\cos\theta\sin\theta & \sin^2\theta - \cos^2\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$$

IV) Elementary Decomposition of Matrix (Imp!!)

eg: $\begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$ Find elementary decomposition | Give geometric interpretation

① First Convert A into I

$$A = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$$

$\downarrow R_1 \to R_1/4$

$$\equiv \begin{bmatrix} 1 & 3/4 \\ 2 & 1 \end{bmatrix}$$

$\downarrow R_2 \to R_2 - 2R_1$

$$\equiv \begin{bmatrix} 1 & 3/4 \\ 0 & -1/2 \end{bmatrix}$$

$\downarrow R_2 \to R_2 \times -2$

$$\equiv \begin{bmatrix} 1 & 3/4 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_1 \to R_1 - 3/4 R_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

② Do individual row transformations on I to get E_i

$R_1 \to R_1/4$

$R_2 \to R_2 - 2R_1$

$R_2 \to -2R_2$

$R_1 \to R_1 - 3/4 R_2$

$$E_1 = \begin{bmatrix} 1/4 & 0 \\ 0 & 1 \end{bmatrix} \quad E_3 = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$$

$$E_2 = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \quad E_4 = \begin{bmatrix} 1 & -3/4 \\ 0 & 1 \end{bmatrix}$$

③ Write in correct order

$$E_4 \cdot E_3 \cdot E_2 \cdot E_1 \overset{A}{\equiv} I$$

$$\therefore A = E_1^{-1} \cdot E_2^{-1} \cdot E_3^{-1} \cdot E_4^{-1} I$$

④ Give geometric interpretation of $(E_i)^{-1}$ matrix

$$A = \underset{\text{scale abt x-axis by } k=4}{E_1^{-1}} \underset{\text{shear abt y-axis by } k=2}{E_2^{-1}} \underset{\text{scale abt y-axis by } k=1/2 \text{ and reflection abt x-axis}(\because -1)}{E_3^{-1}} \underset{\text{shear abt x-axis by } k=3/4}{E_4^{-1}}$$

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & 3/4 \\ 0 & 1 \end{bmatrix}$$

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V) # LU Decomposition

$L \cdot U = A$

L ↑ lower △ , U ↑ upper △

eg: solve $\left.\begin{matrix} x+2y=-1 \\ 3x+4y=-1 \end{matrix}\right\}$ by LU Decomposition

① Write in terms of $Ax=b$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$A \quad x = b$

② Now write A as LU

$$A = \underset{L_0}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} \underset{U_0}{\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}}$$

$\downarrow R_2 \to R_2 + 3R_1$ $\quad \downarrow R_2 \to R_2 - 3R_1$

$$\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}$$

$\Vert\!\Vert$ $\quad \Vert\!\Vert$

$L \quad U$

(i) keep reducing U to its row echelon form to make upper △

(ii) for every transformation done on U, do -ve of it on L

(iii) do elementary operations not row operations

③ $AX = b$

$L\underline{U\,X} = b$

express $\begin{bmatrix} p \\ q \end{bmatrix} = U\begin{bmatrix} x \\ y \end{bmatrix}$

$$\begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} \quad \text{—} (*)$$

④ $L\begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

solve

$$\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \qquad p = -1$$

$3p + q = -1$

$q = -1 + 3 = 2$

⑤ Use p & q to solve eqn (*)

$$\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

$x + 2y = -1$

$-2y = 2$

$\therefore y = -1$

$x + 2(-1) = -1$

$x = -1 + 2$

$x = 1$

$\therefore \boxed{\begin{matrix} x = 1 \\ y = -1 \end{matrix}}$

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VI) PLU Decomposition

[used when first pivot to start LU is missing]

say $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

* $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$R_2 \leftrightarrow R_1$

$P_{12} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (Permutation Matrix)

$P \cdot P^T = I = P^T P$

orthogonal.

$$P_{12} B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

↑ left multiplication — ↑ rows exchanged

$$B P_{12} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

↑ right multiplication — ↑ columns exchanged.

also (det A) = det(PLU)

eg: $x+y+z = 1$

$2x+2y+5z = 0$

$4x+6y+8z = 0$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 5 \\ 4 & 6 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

A X B

$A = LU$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 5 \\ 4 & 6 & 8 \end{bmatrix}$$

$R_1 \to R_2 + 2R_1$, $R_1 \to R_3 + 4R_1$ ↓ $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 4R_1$

$$\approx \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 3 \\ 0 & 2 & 4 \end{bmatrix}$$

Imp ✡✡ ✡✡ since second pivot is zero, we need to use permutation matrix to swap

$R_3 \leftrightarrow R_2$

$$P_{132} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

same as exchange of eqns or making system diagonally dominant.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 2 & 4 \end{bmatrix}$$ left multiplic

$$\therefore \approx \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}}_{\text{right multiplic}^n} P_{132} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 1 \\ 4 & 1 & 0 \end{bmatrix}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \left(P_{132} \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

$Ax = b$

$P_{132} LU x = b$ $\quad [P^{-1} = P]$

$LU x = (P_{132})^{-1} b$

$(LU) X = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

~~solve~~ Let $UX = y$

solve for $Ly = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

sub y into $UX = y$ to get X

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VII) Orthogonal Maps in 2D [Imp Proof]

[either rotation / reflection]

$\boxed{A^TA = AA^T = I \quad \text{i.e } A^T = A^{-1}}$ (orthogonality)

① Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$

② $A \cdot A^T = I$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a^2+b^2 & ac+bd \\ ac+bd & c^2+d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$A^T \cdot A = I$

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a^2+c^2 & ab+cd \\ ab+cd & b^2+d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

③ $\therefore a^2+b^2 = 1$
$ac+bd = 0$
$c^2+d^2 = 1$

$a^2+b^2=1$, $c^2+d^2=1$ } (a, b) & (c, d) lie on same circle

$(a, b) \equiv (\cos\theta, \sin\theta)$ $r=1$

④ $\because ac+bd = 0$

$\langle (a,b), (c,d) \rangle = 0$ [dot product]

$(a,b) \perp (c,d)$

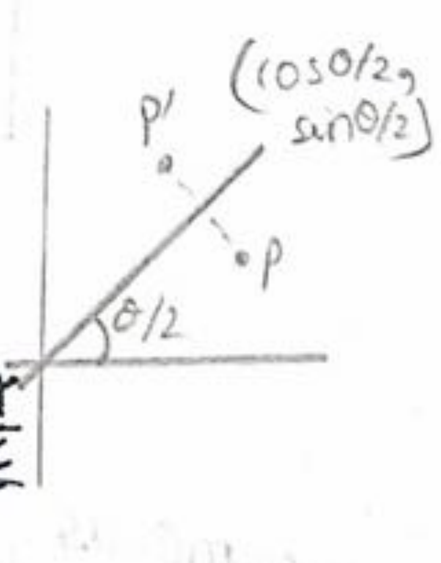
$\therefore (c,d) = \left[\cos\left(\theta+\frac{\pi}{2}\right), \sin\left(\theta+\frac{\pi}{2}\right)\right]$ or $\left[\cos\left(\theta-\frac{\pi}{2}\right), \sin\left(\theta-\frac{\pi}{2}\right)\right]$

$= [-\sin\theta, \cos\theta]$ or $[\sin\theta, -\cos\theta]$

$$\therefore \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \text{ or } \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$$

clockwise rotation matrix

reflection matrix abt line with $\theta' = \theta/2$



$\det = 1 / -1$

$= \pm 1$

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VIII) Rotations in 3D

Rotation abt x-axis or in y-z plane
[ROLL] α

$$\begin{array}{c} \\ x \\ y \\ z \end{array}\begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \end{array} \qquad \begin{array}{c} \\ y \\ z \end{array}\begin{array}{c} \begin{array}{cc} y & z \end{array} \\ \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \end{array}$$

Rotation abt y-axis or in z-x plane
[PITCH] β

$$\begin{array}{c} \\ x \\ y \\ z \end{array}\begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \end{array} \qquad \begin{array}{c} \\ z \\ x \end{array}\begin{array}{c} \begin{array}{cc} z & x \end{array} \\ \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \end{array}$$

Rotation abt z-axis or in x-y plane
[YAW] γ

$$\begin{array}{c} \\ x \\ y \\ z \end{array}\begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \qquad \begin{array}{c} \\ x \\ y \end{array}\begin{array}{c} \begin{array}{cc} x & y \end{array} \\ \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \end{array}$$

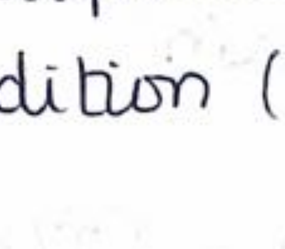
#Euler Angles

$\alpha \xrightarrow{x\text{-axis}}$ ROLL

$\beta \xrightarrow{y\text{-axis}}$ PITCH

$\gamma \xrightarrow{z\text{-axis}}$ YAW

Quarternions



$i = \sqrt{-1}$

$j = \sqrt{-1}$

$k = \sqrt{-1}$

IX) RANK-NULLITY THEOREM

Suppose $T: V_F \longrightarrow W_F$ -- Vector spaces over a field F (scalars)

$\alpha A \in V \rightarrow$ closed under scalar multiplication (scaling)

$A + B \in V \rightarrow$ closed under vector addition (parallelogram law holds)

$U \subseteq V$ is subspace if it is closed under scaling & vector addition

SUBSPACE should pass through the ORIGIN

① Kernel of $T^{\subseteq V}$ / Null Space $\Rightarrow \ker(T) = \{ x \in V \mid T(x) = 0 \}$

② Image of T / Range Space $\Rightarrow Im(T) = \{ T(x) \mid x \in V \}$ $\subseteq W$

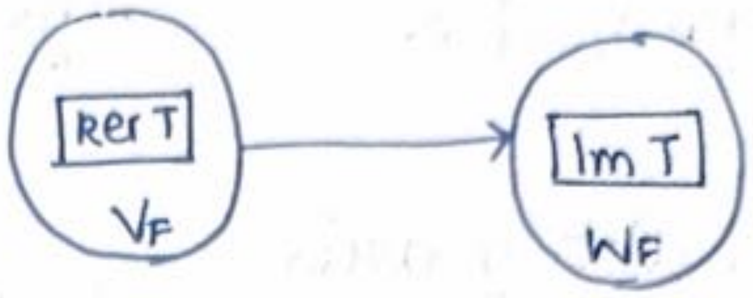
T is a L.T, $\ker(T) \subseteq V$, $im(T) \subseteq$ domain W (domain)

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(S) Let $T: V \to W$ be a Linear Transformation

① Ker(T) is a subspace of domain V

② Img(T) is a subspace of domain W.



Proof:

(A) $Ker(T) \subseteq V$

① Let $x, y \in ker(T) = \{a \in V \mid T(a) = 0\}$

② $\therefore T(x) = 0$
$T(y) = 0$

③ Consider $T(x+y) = T(x) + T(y)$ [$\because$ T is a L-T]
$= 0 + 0$
$= 0$ ——(i)

$x+y \in ker(T)$
kernel is closed under vector addn (11 gm law holds)

④ Let $\alpha \in F$ (scalar

$T(\alpha x) = \alpha T(x)$ [$\because$ T is a L-T]
$= \alpha 0$
$= 0$ ——(ii)

$\alpha x \in ker(T)$
kernel is closed under scalar multiplication

⑤ $\therefore \boxed{Ker(T) \subseteq^{dom} V}$

(B) $Im(T) \subseteq^{codomain} W$

① Let $x, y \in Im(T) = \{T(a) \mid a \in V\}$

② $x = T(a)$, $y = T(b)$ } for some $a, b \in V$

③ $T(a+b) = T(a) + T(b)$
$= x + y$
$\therefore x+y = T(a+b)$
image preimage

$x + y \in Im(T)$ [closed under vector addn]

④ Let $\alpha x = \alpha T(a)$, $a \in V$
$\alpha x = T(\alpha a)$ ($\because$ T is L-T)

⑤ $\alpha x \in Im(T)$ [closed under scalar multiplicn]

⑤ $\therefore \boxed{Im(T) \subseteq^{cod} W}$

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Rank - Nullity Theorem

Let $T: V \to W$ be a linear transformation from finite dimensional vector space V to vector space W such that:

① $\ker(T) = \{ x \in V \mid T(x) = 0 \}$
② $\text{Im}(T) = \{ T(x) \mid x \in V \}$

Denote: $\text{rank}(T) = \dim \text{Im}(T)$

$\text{nullity}(T) = \dim \ker(T)$

Then $\boxed{\text{rank}(T) + \text{nullity}(T) = \dim V}$

Proof: ① We know $\ker(T) \subseteq \text{dom } V$, $\text{Im}(T) \subseteq \text{cod } W$ } already proved.

② Take $U = \{ b_1, b_2, \ldots b_m \}$ as a basis of $\ker(T)$

③ As $\ker T \subseteq V$, we can extend it to

$B = \{ b_1, b_2, \ldots b_m, v_1, v_2 \ldots, v_n \}$

as basis of whole space V

$\therefore \dim V = m + n = n(B)$

Claim: The basis for $\text{Im}(T)$ is : $\{ T(v_1), T(v_2) \ldots, T(v_n) \}$

To prove: Basis for $\text{Im}(T)$ is (i) Linearly Independent (ii) Spanning set for $\text{Im}(T)$

(i) ① Take: ~~$\alpha_1 v_1 + \alpha_2 v_2 + \ldots \alpha_n v_n = 0$~~

$\alpha_1 T(v_1) + \alpha_2 T(v_2) + \ldots \alpha_n T(v_n) = 0$

$\Rightarrow T(\alpha_1 v_1) + T(\alpha_2 v_2) + \ldots T(\alpha_n v_n) = 0$

[$\because$ T is a L.T]

$\Rightarrow T(\alpha_1 v_1 + \alpha_2 v_2 + \ldots \alpha_n v_n) = 0$

② $\therefore \alpha_1 v_1 + \alpha_2 v_2 + \ldots \alpha_n v_n \in \ker(T)$ [$\because T(x) = 0 \mid x \in V$]

③ if $T(x) = 0$, x will belong to $\ker T$

U is basis for $\ker T$

$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \ldots \alpha_n v_n = \beta_1 b_1 + \beta_2 b_2 + \ldots \beta_m b_m$

$\alpha_1 v_1 + \alpha_2 v_2 + \ldots \alpha_n v_n - \beta_1 b_1 - \beta_2 b_2 \ldots - \beta_m b_m = 0$

④ $\alpha_i = 0 \quad \forall i = 1, 2 \ldots, n$

$\beta_i = 0 \quad \forall i \in 1, 2, \ldots m$ ($\because$ B is basis for V)

⑤ $\therefore \{ T(v_1), T(v_2), \ldots T(v_n) \}$ is LINEARLY INDEPENDENT.

(ii) ① Take $y \in \text{Im}(T)$

② $y = T(x)$ for some $x \in V$

③ $x = \alpha_1 b_1 + \alpha_2 b_2 + \ldots \alpha_m b_m + \beta_1 v_1 + \beta_2 v_2 + \ldots \beta_n v_n$

④ $T(x) = \alpha_1 T(b_1) + \ldots \alpha_m T(b_m) + \beta_1 T(v_1) + \ldots \beta_n T(v_n)$

$= \alpha_1 (0) + \ldots \alpha_m (0) + \beta_1 T(v_1) + \ldots \beta_n T(v_n)$

⑤ $\therefore y = T(x) = \beta_1 T(v_1) + \beta_2 T(v_2) + \ldots \beta_n T(v_n)$

$\therefore$ spanning set proved $\therefore$ basis $\text{Im}(T)$ proved

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⊛ Differentiation $(T : P_n \to P_{n-1})$

① Linear Transformation

$$T(f) = f'$$

$$T(f+g) = T(f) + T(g)$$
$$= f' + g'$$
$$= \frac{df}{dx} + \frac{dg}{dx}$$
$$= \frac{d}{dx}(f+g) = (f+g)'$$

$$T(\alpha f) = (\alpha f)'$$
$$= \frac{d}{dx}(\alpha f)$$
$$= \alpha \frac{df}{dx}$$
$$= \alpha f'$$
$$T(\alpha f) = \alpha T(f)$$

$\therefore$ $T(f)$ is linear transformation

② Rank Nullity

(i) $\ker(T) = \{ f \in P_n \mid T(f) = 0 \}$

$$T(f) = 0 \Rightarrow f' = 0$$
$$= \{ a_0 + a_1 x + a_2 x^2 + \dots a_n x^n \mid a_1 + 2a_2 x + \dots n a_n x^{n-1} = 0 \}$$

$$\therefore a_1 = 0$$
$$a_2 = 0$$
$$\vdots$$
$$a_n = 0$$

$\therefore \ker T = \{ a_0 \mid a_0 \in \mathbb{R} \}$

Basis for $\ker(T) = n(B) = 1$

(ii) $\text{Im}(T) = \{ T(f) \mid f \in P_n \}$
$$= \{ a_1 + 2a_2 x + 3a_3 x^2 + \dots n a_n x^{n-1} \mid a_i \in \mathbb{R} \}$$

Basis for $\text{Im}(T) = B = \{ 1, x, x^2 \dots, x^{n-1} \}$

$$\dim(\text{Im}\, T) = n - 1 + 1 = n$$

$$\therefore \dim(P_n) = n+1 = \dim(\text{Im}) + \dim(\ker T)$$
$$= n + 1$$

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③ matrix form of $T(f) = f'$

Consider $f = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \mid a_i \in R$

$T: P_3 \to P_2$

Basis for domain $= \{1, x, x^2, x^3\}$

Basis for codomain $= \{1, x, x^2\}$

$T(1) = 0 = 0(1) + 0(x) + 0(x^2) + \cancel{0(x^3)}$

$T(x) = 1 = 1(1) + 0(x) + 0(x^2)$

$T(x^2) = 2x = 0(1) + 2(x) + 0(x^2)$

$T(x^3) = 3x^2 = 0(1) + 0(x) + 3(x^2)$

$$m(T) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}^T = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$ corresponds to derivative.

$\cdots n$

⊛ Integration $(T: P_n \to P_{n+1})$

① Linear Transformation.

$T(f) = \int f \, dx$

$T(f+g) = \int f + g \, dx$

$= \int f \, dx + \int g \, dx$

$= T(f) + T(g)$

$T(\alpha f) = \int \alpha f \, dx$

$= \alpha \int f \, dx$

$= \alpha \, T(f)$

$\alpha \to$ scalar.

$\therefore$ Linear Transformation

② Rank - Nullity

(i) $\ker(T) = \{ f \in P_{n+1} \mid T(f) = 0 \}$

$T(f) = 0 \Rightarrow \int f \, dx = 0$

$= \{ a_0 + a_1 x + a_2 x^2 + \ldots a_n x^n \mid a_0 x + \frac{a_1 x^2}{2} + \ldots \frac{a_n x^{n+1}}{n+1} + C = 0 \}$

$\therefore a_0 = 0$

$a_1 = 0 \ldots a_n = 0$

$\therefore C = 0$

$\ker(T) = \{0\}$

$\Rightarrow \dim(\ker T) = 0$

(ii) $\mathrm{Im}(T) = \{ T(f) \mid f \in P_n \}$

$\left\{ a_0 x + \frac{a_1 x^2}{2} + \frac{a_2 x^3}{3} + \ldots \frac{a_n x^{n+1}}{n+1} + C \;\middle|\; a_i \in R \right\}$

Basis for $\mathrm{Im}(T) = \{ x, x^2 \ldots x^{n+1} \}$

$n(B) = n + 1$

$\therefore \dim(V) = \dim(\ker) + \dim(\mathrm{Im})$

$n+1 = 0 + n + 1$

proved verified

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③ Matrix corr. to Integration.

Consider $P_2 \to P_3$

Basis for $P_2 = \{1, x, x^2\}$

Basis for $P_3 = \{1, x, x^2, x^3\}$

$T(1) = \int 1\,dx = x + C_1 = (1)(C_1) + 0(x) + 0(x^2) + 0(x^3)$

$T(x) = \int x\,dx = \frac{x^2}{2} + C_2 = (1)(C_2) + 0(x) + \frac{1}{2}(x^2) + 0(x^3)$

$T(x^2) = \int x^2\,dx = \frac{x^3}{3} + C_3 = (1)(C_3) + 0(x) + 0(x^2) + \frac{1}{3}(x^3)$

$T(x^3) = \int x^3\,dx = \frac{x^4}{4} + C_4 = (1)(C_4) + 0(x) + 0(x^2) + 0(x^3) + \frac{1}{4}(x^4)$

$$m(T) = \begin{bmatrix} C_1 & 1 & 0 & 0 \\ C_2 & 0 & 1/2 & 0 \\ C_3 & 0 & 0 & 1/3 \end{bmatrix}^T = \begin{bmatrix} C_1 & C_2 & C_3 \\ 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}$$

antiderivative matrix.

* if $\ker(T) = 0$, dimensions as it is
* if $\ker(T) = x$, dimension reduced by x

Functions + R·N·T.

Result 1:

Function f is one-one if $f(x) = f(y) \Rightarrow x = y$

(images coincide) $\Rightarrow$ (pre images coincide)

$T: V \to W$ is a L·T

Then "T" is one-one iff $\ker(T) = \{0\}$

$\to$ i.e T is one-one $\iff \ker(T) = \{0\}$.

To prove: ($\Rightarrow$ assume T is one-one, prove $\ker(T) = 0$ using $0 \subseteq \ker$, $\ker \subseteq 0$

$\Leftarrow$ assume $\ker(T) = \{0\}$, prove $T(x) = T(y) \Rightarrow x = y$)

Result 2: $T: V \longrightarrow W$ is a L·T.

$Im(T) = \{T(x) \mid x \in V\} \subseteq W$

Then "T" is ONTO iff $Im(T) = W$

To prove: ($\Rightarrow$ Assume T is onto, prove $Im(T) = W$ (Take $u \in W$, $\exists\, x \in V$ s.t $T(x) = u$, $u \in Im(T)$ $\therefore W \subseteq Im(T)$)

$\Leftarrow$ Assume $Im(T) = W$, prove T is onto.
Take $y \in W \Rightarrow y \in Im(T) \Rightarrow y = T(x)$ $\forall\, x \in V$ $\therefore$ T is onto)

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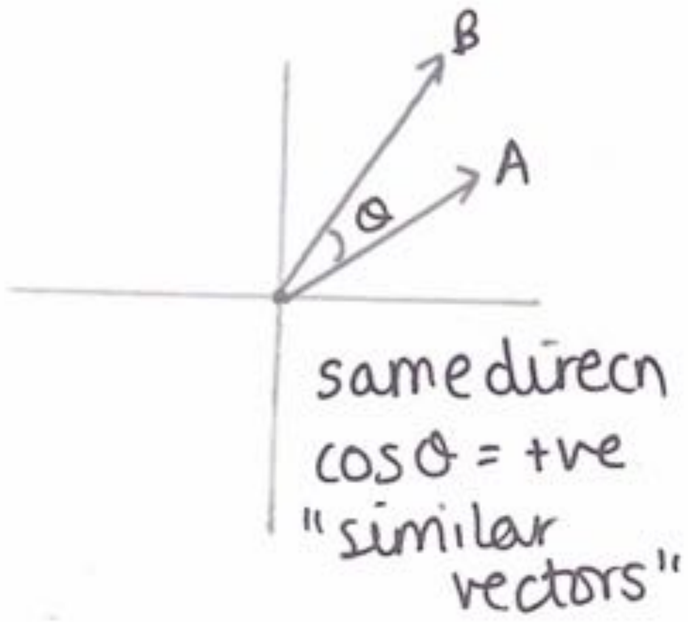
$\therefore$ T is one-one $\iff$ $\ker(T) = \{0\}$
T is onto $\iff$ $\text{Im}(T) = W$

Let T: V → W be a L·T
Let T is one-one
Let T is onto
} then $\boxed{\dim W = \dim V}$
then $\boxed{V \cong W}$ V isomorphic W to

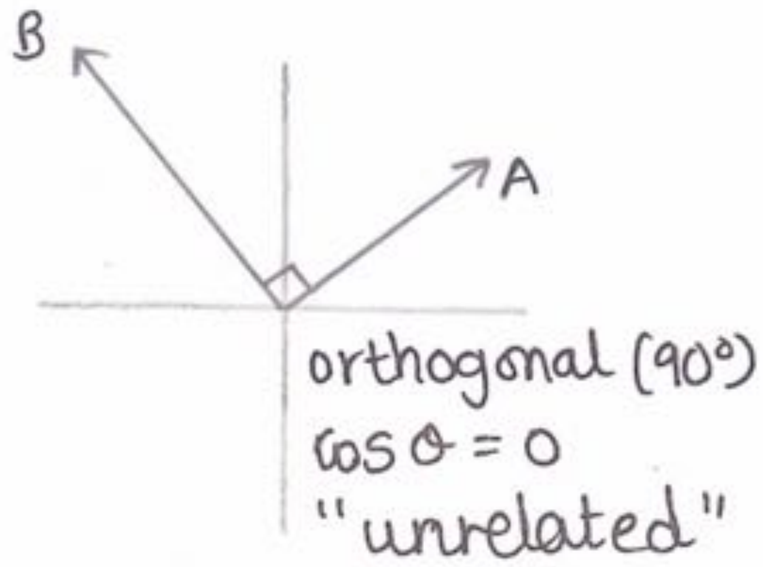
→ ① T is L·T
→ ② T is one-one.
→ ③ T is onto.

* prove these to prove that T its is isomorphic.

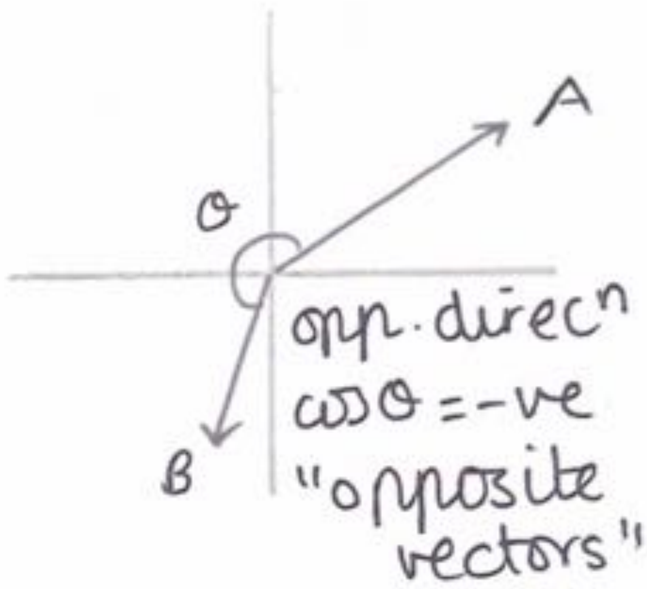
X) COSINE SIMILARITY + PROJECTION MATRIX



same direcn
$\cos\theta$ = +ve
"similar vectors"



orthogonal (90°)
$\cos\theta = 0$
"unrelated"



opp. direcn
$\cos\theta$ = -ve
"opposite vectors"

* Law of Cosines : $c^2 = a^2 + b^2 - 2ab\cos\theta$

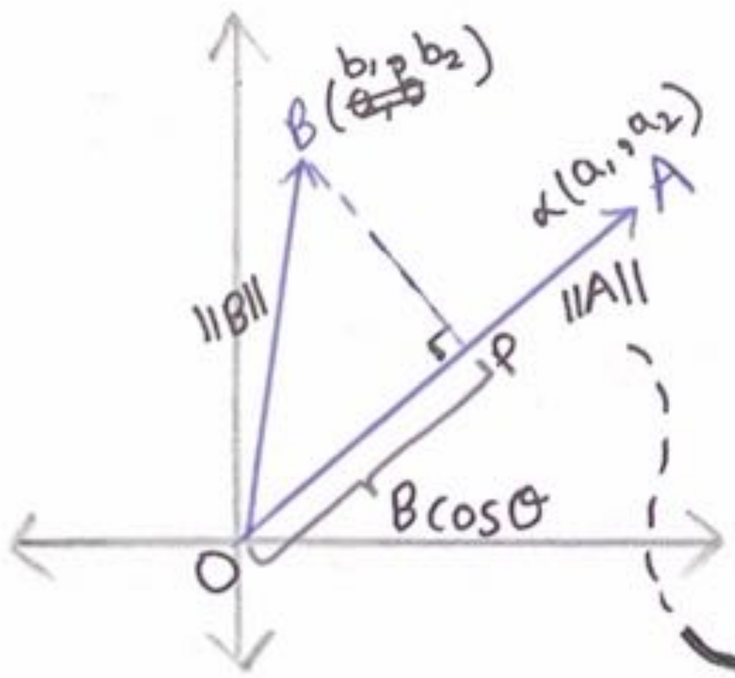
* Dot product : $A = (a_1, a_2 \ldots, a_n)$, $B = (b_1, b_2 \ldots, b_n)$ } $A \cdot B = \langle A, B \rangle = a_1b_1 + a_2b_2 + \ldots + a_nb_n$

$\|A\| \Rightarrow$ dist from origin.

$A \cdot A = \langle A \cdot A \rangle = \|A\|^2 = a_1^2 + a_2^2 + \ldots a_n^2$

* $\boxed{A \cdot B = \|A\| \|B\| \cos\theta}$

$\boxed{\cos\theta = \dfrac{A \cdot B}{\|A\| \|B\|} = \text{sim}(A, B)}$ cosine similarity formula



unit vector

$\boxed{\text{Proj}_A(B) = \dfrac{\langle A, B \rangle}{\|A\|} \hat{A}}$ projection of B on A

$\boxed{\text{Proj}_A(B) = \dfrac{\langle A, B \rangle}{\|A\|}}$ scalar distance/ length of projection of B on A

another method to find : $OP \perp PB$
$\langle ((b_1 - \alpha)(b_2 - \alpha)), T\langle(a_1, a_2)\rangle = 0$

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QR Decomposition

- useful for inconsistent systems
- gives solution without matrix inverse

(*) Gram Schmidt —problem→ Numerical Errors —solution→ LENGTH-INVARIANT OPERATIONS (helps handle numerical errors)

① Rotation [Given's Matrices]
② Reflection [Householder's Method]

Given's Matrices

① y-z plane (rotation in y-z plane)

$$\begin{matrix} & x & y & z \\ x \\ y \\ z \end{matrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix}\begin{matrix} \\ ② \\ ③ \end{matrix}$$

② x-y plane (rotation in x-y plane)

$$\begin{matrix} \\ ① \\ ② \end{matrix}\begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

③ z-x plane (rotation in x-z plane)

$$\begin{bmatrix} c & 0 & -s \\ 0 & 1 & 0 \\ s & 0 & c \end{bmatrix}\begin{matrix} ① \\ \\ ③ \end{matrix}$$

(different from normal rotation matrix)

$\cos\theta = c$, $\sin\theta = s$ } θ need not be calculated

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} cr \\ -sr \end{bmatrix}$$

$\boxed{c = x/r \quad s = -y/r}$

$A_{m\times n} = Q_{m\times n} R_{n\times n}$ ← upper Δ

Q: orthonormal

$\boxed{G^{-1} = G^T}$ ★

eg: $A = \begin{bmatrix} 3 & 2 & 0 \\ 4 & 11 & 5 \\ 0 & 12 & 2 \end{bmatrix}$ (3: x, 4: y)

① Apply Given's Rotation on x and y as $\begin{bmatrix} c & -s \\ s & c \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix}$

$x = 3$ $\quad r = \sqrt{x^2+y^2} = 5$

$y = 4$ $\quad c = \frac{x}{r} = \frac{3}{5}$

$s = \frac{-y}{r} = \frac{-4}{5}$

operation on R_1, R_2

$$G_1 A = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 3 & 2 & 0 \\ 4 & 11 & 5 \\ 0 & 12 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3/5 & +4/5 & 0 \\ -4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 3 & 2 & 0 \\ 4 & 11 & 5 \\ 0 & 12 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 10 & 4 \\ 0 & 5 & 3 \\ 0 & 12 & 2 \end{bmatrix}$$

② Again apply Given's Rotation on new x & y

$$\begin{bmatrix} 5 & 10 & 4 \\ 0 & 5 & 3 \\ 0 & 12 & 2 \end{bmatrix}$$

$x = 5$ $\quad c = 5/13$

$y = 12$ $\quad y = -12/13$

$r = 13$

operation on R_2, R_3

$$G_2 \begin{bmatrix} 5 & 10 & 4 \\ 0 & 5 & 3 \\ 0 & 12 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5/13 & 12/13 \\ 0 & -12/13 & 5/13 \end{bmatrix}\begin{bmatrix} 5 & 10 & 4 \\ 0 & 5 & 3 \\ 0 & 12 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 10 & 4 \\ 0 & 13 & 3 \\ 0 & 0 & -2 \end{bmatrix} = R$$

③ $G_2 . G_1 . A = R$ [map out the transform^n's done to get R and rearrange as QR]

$A = (G_1^{-1} G_2^{-1}) R$

$= \boxed{G_1^T G_2^T} R$ (as G is orthogonal)

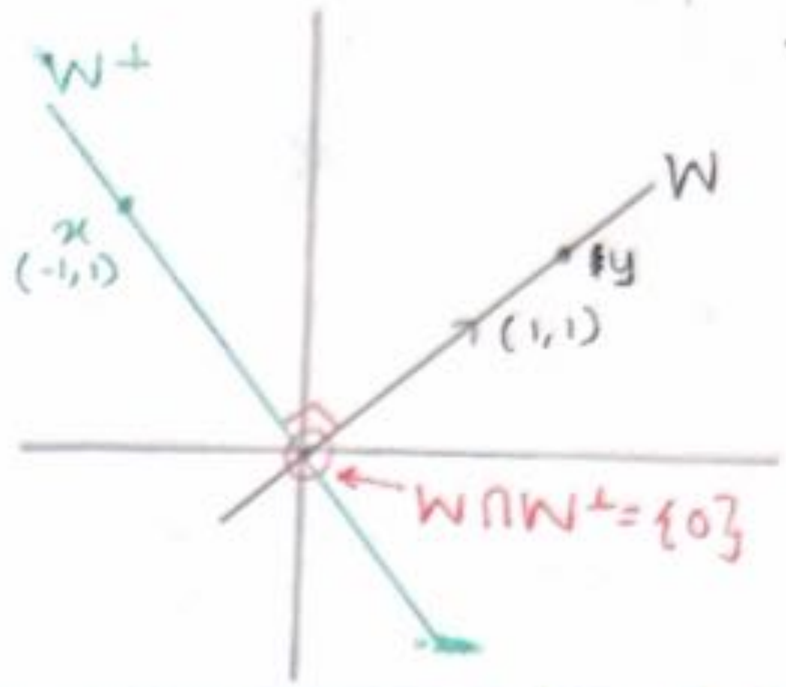
Q

④ calculate Q

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ORTHOGONAL COMPLEMENT OF SUBSPACE.

$W^\perp = \{x \in V \mid \langle x, y\rangle = 0\}$ $\forall y \in W$, $W \subseteq V$



Result 1: $W \cap W^\perp = \{0\}$

Proof: ① Take $t \in W \cap W^\perp$

$\therefore t \in W$ and $t \in W^\perp$

$\therefore \langle t, t\rangle = 0$

$\therefore \boxed{t = 0}$

$\Rightarrow W \cap W^\perp \subseteq \{0\}$ —①

② Take $y \in \{0\}$

$y = 0$ falls in W

$\therefore y \in W$ ($\because$ W is a subspace of V and every subspace contains 0)

Now,

$\langle 0, x\rangle = 0 \quad \forall x \in W$

$\Rightarrow 0 \in W^\perp$

$\Rightarrow 0 \in W \cap W^\perp$ (as 0 belongs to W and $W^\perp$)

$\Rightarrow \{0\} \subseteq W \cap W^\perp$ —②

From ① & ②, $W \cap W^\perp = \{0\}$

Result 2: $W^\perp$ is a subspace of V

① Take $x, y \in W^\perp$

$\langle x, t\rangle = 0 \quad \forall t \in W$

$\langle y, t\rangle = 0$

② Consider $\langle x+y, t\rangle$

$= \langle x+y, t\rangle$

$= \langle x, t\rangle + \langle y, t\rangle$

$= 0 + 0$

$= 0$ $\therefore x+y \in W^\perp$

③ Consider $\langle \alpha x, t\rangle$

$= \alpha \langle x, t\rangle$

$= \alpha . 0$

$= 0$ $\therefore \alpha x \in W^\perp$

by ② & ③, $W^\perp$ is closed under addition & scalar multiplication

$\therefore W^\perp$ is a subspace of V

Result 3: Take any subspace W of V. Then $V = W \oplus W^\perp$

Proof: ① Let P be the projection matrix corresponding to projection on column space of W.

② Take $x \in V$

$x = \underline{Px} + \underline{x - Px}$

Px = projec^n of x on col. space of W

$\therefore Px \in W$

To prove: $x - Px \in W^\perp$

③ $\langle t, x - Px\rangle \quad (t \in W)$

$= \langle Pt, x - Px\rangle$ ($\because$ $t \in W$, projecn of t $\in W$)

$= (Pt)^T (x - Px)$ (vectorisation)

$= t^T P^T (x - Px)$ $[P^T = P, P^2 = P]$

$= t^T P (x - Px)$

$= t^T (Px - P^2 x)$

$= t^T (Px - Px)$

$= t^T (0)$

$= 0$

$\therefore \langle t, x - Px\rangle = 0$

④ $x - Px \in W^\perp$ and $W \cap W^\perp = 0$

$\therefore \boxed{V = W \oplus W^\perp}$

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Note: Direct Sum of Subspaces

⇒ If $V = V_1 + V_2$; V_1 & V_2 are subspaces of V

$V_1 \cap V_2 = \{0\}$

⇒ Then V is called direct sum of V_1, V_2 $\Rightarrow V_1 \oplus V_2$

Result : Let U, W be subspaces of V (vector space)

Then ⇒ $\dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W)$

[Proof check notebook]

Corollary: ① $\dim(U \oplus W) = \dim U + \dim W$

② $V = W \oplus W^\perp$

$\dim V = \dim W + \dim W^\perp$ ($\because W \cap W^\perp = \{0\}$)

FOUR FUNDAMENTAL SUBSPACES

$T: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ $\quad T(X) = A_{m\times n} X_{n\times 1}$ $\quad T^*: \mathbb{R}^m \longrightarrow \mathbb{R}^n$

$x \longrightarrow Ax \quad T^*(X) = A^T_{n\times m} X_{m\times 1}$

① $\underline{\ker(T)} = \{x \in \mathbb{R}^n \mid T(x) = 0\}$
$= \{x \in \mathbb{R}^n \mid AX = 0\}$
$= \underline{N(A)}$ null space of A

$\underline{\ker(T^*)} = \{T^*(x) = 0 \mid x \in \mathbb{R}^m\}$
$= \{x \in \mathbb{R}^m \mid A^T X = 0\}$
$= \underline{N(A^T)}$ null space of A^T.

② $T(x) = AX$

$\underline{C(A)} = \{Ax \mid x \in \mathbb{R}^n\}$
$= \{x_1 c_1 + x_2 c_2 + \ldots x_n c_n\}$
$= \{$L.C's of columns of $A\}$

$A = [c_1\ c_2 \ldots c_n]$ $\quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

$C(A)$ = span of columns of A

↳ subspace of $\mathbb{R}^m$

$\mathbb{R}^m = C(A) + C(A)^{\perp}$

any vector space is a direct sum of a subspace and its orthogonal complement.

$C(A)^\perp = N(A^T)$

① Take $y \in C(A^T)$

$\Leftrightarrow \langle Y, AX \rangle = 0 \quad \forall X \in \mathbb{R}^n$ (domain)
(inner product in $\mathbb{R}^m$)

$\Leftrightarrow Y^T . AX = 0$ (vectorization).

$\Leftrightarrow (Y^T A) X = 0$ (associativity)

$\Leftrightarrow (A^T Y)^T X = 0 \quad \forall x \in \mathbb{R}^n$

$\Leftrightarrow \langle A^T Y, X \rangle = 0 \quad \forall X \in \mathbb{R}^n$.
Inner product in $\mathbb{R}^n$

$\Leftrightarrow A^T Y$ is perpendicular to every vector X in $\mathbb{R}^n$

$\Leftrightarrow A^T Y = 0$

$Y \in N(A^T)$

$\therefore C(A)^\perp = N(A^T)$

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$T: \mathbb{R}^n \longrightarrow \mathbb{R}^m$

$T(X) = A_{m\times n} X_{n\times 1}$

$C(A^T) \oplus N(A) = \mathbb{R}^n$

$R(A) \oplus N(A) = \mathbb{R}^n$

$T^*: \mathbb{R}^m \longrightarrow \mathbb{R}^n$

$T^*(X) = A^T_{n\times m} Y_{m\times 1}$

$C(A) \oplus N(A^T) = \mathbb{R}^m$

$R(A^T) \oplus N(A^T) = \mathbb{R}^m$

$\mathbb{R} : C(A) + C(A^*)^\perp$

$C(A), C(A^T)$

$N(A), N(A^T)$

four fundamental subspaces

$R(A) \Leftrightarrow C(A^T)$

$R(A^T) \Leftrightarrow C(A)$.

range space / row space.

$C(A)^\perp = N(A^T)$

$C(A^T)^\perp = N(A)$

Theorem: Let A be a mxn matrix

Then row rank of A = column rank of A.

Proof: ① If $A = 0$, then row rank of A = column rank of A = 0

② Let $A \neq 0$,

Take the smallest number 'r'

such that: $A_{m\times n} = B_{m\times r} \cdot C_{r\times n}$ [C is in echelon form]

③ row rank (A)

= minimum no. of rows in C that are non-redundant

= r

④ column rank (A)

= minimum no. of columns in B that are non-redundant

= r

⑤ ∴ By ③ & ④, row rank(A) = col. rank(A) = rank(A) = r.

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Tensors

eg: $X = \left\{ \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}_{3\times4}, \begin{bmatrix} 11 & 12 & 13 & 14 \\ 15 & 16 & 17 & 18 \\ 19 & 20 & 21 & 22 \end{bmatrix}_{3\times4} \right\}$

matrices kept on top of each other
(i=1) ⇒ TOP
(i=2) ⇒ NEXT



i = (no. of matrices) $\in \{1,2\}$
j = (no. of rows) $\in \{1,2,3\}$
k = (no. of columns) $\in \{1,2,3,4\}$

$X_{ijk} \Rightarrow$ tensors

Vector space under consideration $\Rightarrow \mathbb{R}^{2\times3\times4}$

① Horizontal Slices $\Rightarrow X(i,:,:) = X$
② Lateral Slices $\Rightarrow X(:,j,:)$
③ Frontal Slices $\Rightarrow X(:,:,k)$

eg: $X(:,j,:) \Rightarrow \left\{ \underset{j=1}{\begin{bmatrix} 1 & 2 & 3 & 4 \\ 11 & 12 & 13 & 14 \end{bmatrix}}, \underset{j=2}{\begin{bmatrix} 5 & 6 & 7 & 8 \\ 15 & 16 & 17 & 18 \end{bmatrix}}, \underset{j=3}{\begin{bmatrix} 9 & 10 & 11 & 12 \\ 19 & 20 & 21 & 22 \end{bmatrix}} \right\}$

all matrices take rows
iterate over rows

$X(:,:,k) \Rightarrow \left\{ \underset{k=1}{\begin{bmatrix} 1 & 5 & 9 \\ 11 & 15 & 19 \end{bmatrix}}, \underset{k=2}{\begin{bmatrix} 2 & 6 & 10 \\ 12 & 16 & 20 \end{bmatrix}}, \underset{k=3}{\begin{bmatrix} 3 & 7 & 11 \\ 13 & 17 & 21 \end{bmatrix}}, \underset{k=4}{\begin{bmatrix} 4 & 8 & 12 \\ 14 & 18 & 22 \end{bmatrix}} \right\}$

all matrices take columns
iterate over cols

$X(1,2,3) \Rightarrow$ 1st matrix, 2nd row, 3rd column $\Rightarrow \boxed{7}$

⊛ Four Fundamental Subspaces Problem + Diagram in CLA IA 4 (visit)

⊛ Note:- If $A = A^T$, then 4 fundamental subspaces $\xrightarrow{\text{reduce to}}$ just 2 fundamental subspaces

$N(A) = N(A^T)$
$R(A) = C(A)$

$\mathbb{R}^n = C(A) + N(A)$ [$T: \mathbb{R}^n \to \mathbb{R}^n$, $T(x) = Ax = b$]

↓ gives direcns which explain b [features from A which explain b] SIGNAL

↘ has direcns which have no influence on 'b' [NOISE/ERROR]

Any $X \in \mathbb{R}^n$ (data) $\Rightarrow X = X_r + X_n$ $\quad X_r \in C(A)$, $X_n \in N(A)$

$AX = b$

① unique solution ⇒ if $N(A) = \{0\}$ ← zero error
i.e full rank and $b \in C(A)$

② ∞-many solutions ⇒ if $\dim(N(A)) > 1$
i.e $\text{rank}(A) < n$
system is consistent
$Ax = b$, $b \in C(A)$

③ no solution ⇒ $b \notin C(A)$
i.e $\text{rank}(A) \neq \text{rank}[A|b]$
inconsistent system
(we used to find least square soln when $r(A) = n$)

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All Proofs After Midsem

① Product of two orthogonal matrices is orthogonal.

① Let A and B be two orthogonal matrices

② To prove orthogonality, we need to prove $A^TA = AA^T = I$

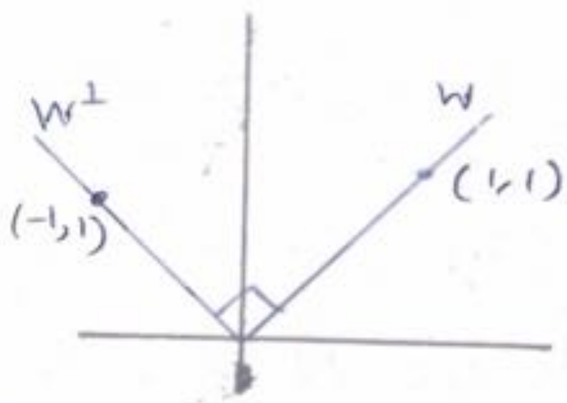
③ $(AB)(AB)^T = ABB^TA^T$
$= AIA^T$
$= AA^T$
$= I$

④ $(AB)^T(AB) = B^TA^TAB$
$= B^TIB$
$= B^TB$
$= I$

[$\because A^TA = AA^T = I$
$B^TA = BB^T = I$]

② Orthogonal complement of subspace

$W^\perp = \{x \in V \mid \langle x, y \rangle = 0 \ \forall \ y \in W\}$



Result 1: $W^\perp \cap W = \{0\}$

① Take $t \in W \cap W^\perp$

$\therefore t \in W$ and $t \in W^\perp$

$\therefore \langle t, t \rangle = 0$

$t = 0$

$\Rightarrow W \cap W^\perp \subseteq \{0\}$

② Take $y = \{0\}$

$\therefore y = 0$ falls in W

$\therefore y \in W$ [W is a subspace of V and every subspace contains 0]

Now $\langle 0, x \rangle = 0 \ \forall \ x \in W$

$0 \in W^\perp$

$0 \in W \cap W^\perp$ (as 0 belongs to W and $W^\perp$)

$\Rightarrow \{0\} \subseteq W \cap W^\perp$

From ① & ② $\boxed{W \cap W^\perp = \{0\}}$

Result 2:

① W⊥ is a subspace of V

① Take $\left.\begin{matrix}\langle x, t \rangle = 0 \\ \langle y, t \rangle = 0\end{matrix}\right\} \forall t \in W$ $x, y \in W^\perp$

② Take $\langle x+y, t \rangle$
$= \langle x, t \rangle + \langle y, t \rangle$
$= 0 + 0$
$= 0 \quad \therefore x+y \in W^\perp$

③ Consider $\alpha x \in W^\perp$ (α some scalar)
$\langle \alpha x, t \rangle$
$= \alpha \langle x, t \rangle$
$= \alpha(0) = 0 \quad \therefore \alpha x \in W^\perp$

④ from ② & ③,
W⊥ is closed under ~~scalar~~ scalar addition & multiplication

$\therefore$ ⑤ $\boxed{W^\perp \text{ is a subspace of } V}$

Result 3: Take any subspace $W \subseteq V$. Then $\boxed{V = W \oplus W^\perp}$

① Let P be a projection matrix corr to projecn on col. space of W

② Take $x \in V$

③ ~~Px lies in V~~ ③ $Px \in W$

$\therefore x = \underline{Px} + \underline{x - Px}$

④ To prove $x - Px \in W^\perp$

$\langle t, x - Px \rangle$
$= \langle Pt, x - Px \rangle$
$= (Pt)^T(x - Px)$

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③ Take $t \in W$

Then $Pt \in W$

$$= \langle t, x - Px \rangle$$

$$= \langle Pt, x - Px \rangle$$

$$= (Pt)^T (x - Px) \quad \text{[vectorisation]}$$

$$= t^T P^T (x - Px)$$

$$= t^T P^T x - t^T P^T P x$$

$$= t^T P^T x - t^T P^2 x \quad [\because P^T = P,\ P^T P = P^2,\ P^2 = P]$$

$$= t^T (P^T x - Px)$$

$$= t^T (Px - Px)$$

$$= 0$$

$\therefore \langle t, x - Px \rangle = 0$

④ $\therefore x - Px \in W^\perp$

and $W \cap W^\perp = 0$

$\therefore \boxed{V = W \oplus W^\perp}$

direct sum of subspaces.

---

③ Let U, W be subspaces of V (vector space V)

Then, $\dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W)$

① First define $T: U \times W \to V$ by $T(u, w) = u + w$

② Prove that T is a L.T.

③ Write eqn for $\ker T$

④ Prove $\ker T \cong U \cap W$

⑤ Take $\left.\begin{matrix} B_1 \\ B_2 \end{matrix}\right\}$ basis for U, W

⑥ B = basis of $U \times W$ [Prove This] using ① spanning set ② L.I

---

① Define $T: U \times W \to V$

by $T(u, w) = u + w$

Claim: T is a Linear Transformⁿ

i) Take $x = (u_1, w_1)$, $y = (u_2, w_2) \in U \times W$

ii) $T(x+y) = T((u_1, w_1) + (u_2, w_2))$

$= T((u_1 + u_2), (w_1 + w_2))$

$= u_1 + u_2 + w_1 + w_2$

$= (u_1 + w_1) + (u_2 + w_2)$

$= T(u_1, w_1) + T(u_2, w_2)$

$= T(x) + T(y)$

(iii) Take a scalar α,

$T(\alpha x) = T(\alpha(u_1, w_1))$

$= T(\alpha u_1, \alpha w_1)$

$= \alpha u_1 + \alpha w_1$

$= \alpha(u_1 + w_1)$

$= \alpha T(u_1, w_1)$

$= \alpha T(x)$

$\therefore T$ is a Linear Transformation

② $\ker T = \{x \in U \times W \mid T(x) = 0\}$

$= \{(u, w) \in U \times W \mid T(u, w) = 0\}$

$= \{(u, w) \in U \times W \mid u + w = 0\}$

$= \{(u, w) \in U \times W \mid u = -w\}$

$= \{(u, -u) \mid u \in U \cap W\}$

$\because u \in U$ and $u = -w \in W$

$u \in U \cap W$

③ To prove: $\ker T \cong U \cap W$

i.e to show

(i) ϕ is one-one

(ii) ϕ is onto

(iii) ϕ is a L-T.

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Claim: $\ker T \cong U \cap W$

Define $\phi : U \cap W \longrightarrow \ker T$

by $\phi(x) = (x, -x)$.

(i) To show: ϕ is onto

Take $y \in \ker T$

$\Rightarrow y = (u, -u)$ for some $u \in (U \cap W)$

$\Rightarrow y = \phi(u) = (u, -u) = y$

$\therefore \phi(u)$ is onto

(ii) ϕ is one-one

Let $\phi(x) = \phi(y)$

$\Rightarrow (x, -x) = (y, -y)$

$\Rightarrow \therefore x = y$

$\Rightarrow \phi$ is one-one

(iii) ϕ is a L.T Take $x, y \in U \cap W$

(a) $\phi(x+y) = \phi((x,-x)+(y,-y))$

$= \phi((x+y)+(-x+-y))$

$= \phi((x+y), -(x+y))$

$= (x, -x) + (y, -y)$

$= \phi(x) + \phi(y)$

(b) $\phi(\alpha x) = (\alpha x, -\alpha x)$

$= \alpha (x, -x)$

$= \alpha \phi(x)$

$\therefore \phi$ is a L.T

$\therefore \ker T \cong U \cap W$ (isomorphic)

$\therefore \dim(\ker T) \cong \dim(U \cap W)$ ——(i)

(4) By Rank-Nullity Theorem for T,

$\dim(U \times W) = \dim(\ker T) + \dim(\mathrm{Im} T)$

$\Rightarrow \dim(U \cap W) + \dim(U + W) = \dim(U \times W)$ ——(ii)

(5) $\dim U = 1$, $\dim V = 1$, $\dim(U \times W) = \dim(\mathbb{R}^2) = 2$ eg

(5) Take

$B_1 = \{u_1, u_2, u_3 \ldots u_m\}$ be a basis for U

$B_2 = \{w_1, w_2, \ldots w_n\}$ be a basis for W

Claim: $B = \{(u_1,0), (u_2,0), \ldots (u_m,0), \ldots (0,w_1), (0,w_2) \ldots (0,w_n)\}$ is a basis for $U \times W$

ie To show: (i) L.I

(ii) spanning set.

(i) Take: $\alpha_1(u_1,0) + \alpha_2(u_2,0) + \ldots \alpha_m(u_m,0) + \beta_1(0,w_1) + \beta_2(0,w_2) + \ldots \beta_n(0,w_n) = (0,0)$

$\Rightarrow (\alpha_1 u_1 + \alpha_2 u_2 + \ldots \alpha_m u_m), (\beta_1 w_1 + \beta_2 w_2 + \ldots \beta_n w_n) = (0,0)$

$\Rightarrow \alpha_1 u_1 + \alpha_2 u_2 + \ldots \alpha_m u_m = 0$

$\beta_1 w_1 + \beta_2 w_2 + \ldots \beta_n w_n = 0$

$\Rightarrow \alpha_1 = 0, \alpha_2 = 0 \ldots, \alpha_m = 0$

$\beta_1 = 0, \beta_2 = 0, \ldots, \beta_m = 0$

[$\because B_1$ and B_2 are linearly independent (basis)]

$\Rightarrow B$ is L.I

(ii) Take any $x \in U \times W$

$x = (u, w)$ for some $u \in U$, $w \in W$

$\Rightarrow x = (u, w)$

$= (\alpha_1 u_1 + \alpha_2 u_2 + \ldots \alpha_m u_m, \beta_1 u_1 + \beta_2 w_2 + \ldots \beta_n w_n)$

$= \alpha_1(u_1, 0) + \alpha_2(u_2, 0) + \ldots \alpha_m(u_m, 0) + \beta_1(0, w_1) + \beta_2(0, w_2) + \ldots \beta_n(0, w_n)$

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$\therefore \Rightarrow B$ spans $U \times W$

$\therefore \Rightarrow B$ is a basis for $U \times W$

⑥ $\therefore \dim(U \times W) = n(B)$

$= m + n$

$\dim(U \times W) = \dim(U) + \dim(W)$ ← (iii)

Putting (iii) in (ii)

$\dim(U \cap W) + \dim(U+W)$

$= \dim(U) + \dim(W)$

$\Rightarrow \therefore \dim(U+W)$

$= \dim(U) + \dim(W) - \dim(U \cap W)$

COROLLARY

① $\dim(U \oplus W) = \dim U + \dim W$

$(\because U \cap W = \{0\})$

② $\dim V = \dim W + \dim W^{\perp}$

$(V = W \oplus W^{\perp})$

⑤ Theorem:

Let A be a $m \times n$ matrix

Prove that:

row rank of A = column rank of A.

① If $A = O$,

then row rank(A) = col rank(A)

$= 0$.

② Let $A \neq 0$

Take the smallest number r, such that:

$A_{m \times n} = B_{m \times r} \cdot C_{r \times n}$

where C is in echelon form

③ Now,

row rank of A = minimum no. of rows in C that are non-redundant

$= r$.

④ column rank of A

= min. no. of columns in B that are non-redundant

$= r$

⑤ By ③ & ④,

row rank of A = column rank of A $= r$

$= \text{rank}(A)$

④ Result: $(C(A^{\perp}) = N(A^T)$ prove

Proof:

① Take $y \in C(A^{\perp})$

$\Leftrightarrow \langle Y, AX \rangle = 0 \quad (\forall X \in \mathbb{R}^n)$

$\Leftrightarrow Y^T(AX) = 0$ (vectorisation)

$\Leftrightarrow (Y^T A)X = 0$ (associativity)

$\Leftrightarrow (A^T Y)^T X = 0 \quad ((AB)^T = B^T A^T)$

$\Leftrightarrow \langle A^T Y, X \rangle = 0 \quad \forall X \in \mathbb{R}^n$

$\therefore A^T Y$ is perpendicular to every vector X in $\mathbb{R}^n$

$\Leftrightarrow \therefore A^T Y = 0$

$\Leftrightarrow Y \in N(A^T)$

$\therefore \boxed{C(A^{\perp}) = N(A^T)}$

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⑥ Let A_{mxn} be a tall matrix ($m > n$) with no. of independent columns = n

Then prove:

(i) A^TA is invertible.

(ii) Left Inverse of A exists.

Proof:

✱ A^TA is invertible [Claim]

① Form a matrix $B = A^TA$

$B^T = (A^TA)^T = A^T.(A^T)^T = A^TA$

$\therefore$ B is symmetric

② Take T to be linear transformn given by B

Claim: T is one-one & onto

(i) Take $T(X) = 0$

$BX = 0$

$(A^TA)X = 0$

$A^T(AX) = 0$

$\Rightarrow AX \in N(A^T)$ —①

(ii) We know that, $AX \in C(A)$ —②

$\therefore AX \in R(A^T)$ —③

$\Rightarrow AX \in N(A^T) \cap R(A^T)$ (by ① & ②)

$\Rightarrow AX = 0 \quad (\because N(A^T) \cap R(A^T) = \{0\})$

$\Rightarrow X \in N(A)$

(iii) Given cols of A are independent and $AX = 0$

If $X \neq 0$, we get L.C of cols of A = 0 with at least one scalars non-zero.

$\Rightarrow$ cols of A won't be independent (contradiction)

$\Rightarrow \therefore AX = 0$

$\Rightarrow X = 0$ (else cols of A will be dependent)

$\Rightarrow$ T is one-one

[$\because \ker T = \{0\}$]

$\therefore \dim(\ker T) = 0$

To prove: T is onto

By Rank-Nullity Theorem

$\text{rank}(T) + \text{nullity}(T) = n$

$\text{rank}(T) + 0 = n$

$\text{rank}(T) = n$

$\therefore \dim(\text{Im}(T)) = n$

$\Rightarrow$ T is onto

$\therefore$ T is one-one & onto

$\Rightarrow T^{-1}$ exists

$\Rightarrow$ **A^TA is invertible**

✱ Let F be the left inverse of A, Then $FA = I$

(F exists) $\Rightarrow$ Claim ✓

The left inverse of A is given by:

$F = (A^TA)^{-1}A^T$

which exists if $(A^TA)^{-1}$ exists

$\Rightarrow$ As columns of A are independent, by (i) A^TA^{-1} exists

$\therefore$ F exists

Left Inverse exists

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⑦ Let A ~~=~~ $n \times n$ be a symmetric matrix. Then:

(i) All eigenvalues of A are real.

(ii) Eigenvectors corresponding to distinct eigenvalues are orthogonal (i.e they form axes)

Proof:

(i) All eigenvalues of A are real.

① $\langle X, Y \rangle = X^T Y$ (vectorisation)

② $\langle AX, AX \rangle \geq 0$ (axiom)

$\Rightarrow (AX)^T AX \geq 0$ (by ①)

$\Rightarrow (X^T A^T) AX \geq 0$

$\Rightarrow (X^T A^T A) X \geq 0$ (associativity)

$\Rightarrow (A^T A X)^T X \geq 0$

$\Rightarrow \langle A^T A X, X \rangle \geq 0$

$\Rightarrow \langle A^2 X, X \rangle \geq 0$ ($\because A = A^T$).

③ $AX = \lambda X$

$A^2 X = \lambda A(AX)$

$= A \lambda X$

$= \lambda AX$

$A^2 X = \lambda^2 X$

$\hookrightarrow$ eigenvalue of A^2

④ By ②,

$\langle A^2 X, X \rangle \geq 0$

$\Rightarrow \langle \lambda^2 X, X \rangle \geq 0$

$\Rightarrow \lambda^2 \langle X, X \rangle \geq 0$

$\Rightarrow \lambda^2 \geq 0$ ($\because \langle X, X \rangle \geq 0$).

⑤ If possible, let $\lambda \in \mathbb{C}$ complex.

$\lambda = a + bi$ (where a, b are real)

$\lambda^2 = (a+bi)(a+bi)$

$= a^2 + abi + abi + (bi)^2$

$= a^2 + 2abi - b^2$

$= (a^2 - b^2) + 2abi.$

$\lambda^2 = (a^2 - b^2) + 2abi \geq 0$

$\therefore ab = 0$

$\Rightarrow a = 0$ or $b = 0$

If $b = 0$, $\lambda = a$ (real).

If $a = 0$, $\lambda = bi$

$\Rightarrow \lambda^2 = (bi)^2$

$\Rightarrow \lambda^2 = -b^2$

$\Rightarrow \lambda^2 \geq 0$

$-b^2 \geq 0$

$\therefore \boxed{b = 0}$ is real.

$\lambda = 0$ is real

$\therefore$ Symmetric matrix has all its eigenvalues as real nos.

(ii) Take $\lambda_1 \neq \lambda_2$ (distinct)

$AX = \lambda_1 X$

$AY = \lambda_2 Y$

To show $\langle X, Y \rangle = 0$

$\langle AX, Y \rangle$

$\Rightarrow (AX)^T Y$

$\Rightarrow X^T A^T Y$

$\Rightarrow$

① $\langle AX, Y \rangle = \langle \lambda_1 X, Y \rangle = \lambda_1 \langle X, Y \rangle = \lambda_1 X^T Y$

② $\langle AX, Y \rangle = \langle A^T X, Y \rangle$ ($\because A = A^T$)

$= (A^T X)^T Y$

$= X^T AY = \langle X, AY \rangle = \langle X, \lambda_2 Y \rangle = \lambda_2 \langle X, Y \rangle$

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$\therefore \langle AX, Y \rangle = \lambda_1 X^T Y$

$= \lambda_2 X^T Y$

$\lambda_1 X^T Y - \lambda_2 X^T Y = 0$

$(\lambda_1 - \lambda_2) X^T Y = 0$

$\lambda_1 \neq \lambda_2 \Rightarrow \therefore \lambda_1 - \lambda_2 \neq 0$

$\therefore X^T Y = 0$

$\therefore \boxed{X \perp Y}$

⑧ Spectral Theorem

$A_{n \times n} = \lambda_1 U_1 U_1^T + \lambda_2 U_2 U_2^T + \dots \lambda_n U_n U_n^T$

Proof:

① Take a symmetric matrix $A_{n \times n}$
Then $\exists$ an orthogonal matrix
P s.t $A = PDP^T$

② $A_{n \times n} \Rightarrow$ symmetric
(i) all eigenvalues are real
(ii) eigenvectors of distinct eigen values are orthogonal

③ If eigenvalues are not distinct, we get orthonormal basis using Gram Schmidt Orthogonalisation

④ Let $\{U_1, U_2, \dots U_n\}$ be unit eigenvectors corr. to A or Gram Schmidt

⑤ Choose $P = [U_1 \; U_2 \dots U_n]$ where
$\langle u_i, u_j \rangle = 0$
$\langle u_i, u_i \rangle = 1$

⑥ $AP = A[U_1 \; U_2 \dots U_n]$

$= [AU_1 \; AU_2 \dots AU_n]$

$= [\lambda_1 U_1 \; \lambda_2 U_2 \dots \lambda_n U_n]$

$= [U_1 \dots U_n] \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{bmatrix}$

$= PD$

⑦ $AP = PD$

$A = PDP^T$

$A = [U_1 \; U_2 \dots U_n] \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ & \lambda_3 & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} U_1^T \\ U_2^T \\ \vdots \\ U_n^T \end{bmatrix}$

EXPLICIT SPECTRAL THEOREM

$A = \lambda_1 U_1 U_1^T + \lambda_2 U_2 U_2^T + \dots \lambda_n U_n U_n^T$

⑨ Principal Axes Theorem

Let $A_{m \times n}$ be a symmetric matrix. Then there exists an orthogonal change of variable $X = PY$ that transforms quadratic form $X^T A X$ into quadratic form $Y^T D Y$ with no cross product terms.

Proof:

① $q(X) = X^T A X$

$= X^T (PDP^{-1}) X$

$= X^T (PDP^T) X$

$= (X^T P) D (P^T X)$

$= Y^T D Y$ where $Y = P^T X$

② $Y = P^T X$

$\therefore X = (P^T)^{-1} Y$

$X = PY \quad (\because PP^T = P^T P = I)$

③ $q(X) = q(Y)$

$= Y^T D Y$

$= \begin{bmatrix} y_1 \\ y_2 \\ y_n \end{bmatrix}^T \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_n \end{bmatrix}$

no cross product terms

$= [y_1 \; y_2 \dots y_n] \begin{bmatrix} \lambda_1 \dots \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_n \end{bmatrix}$

$= \boxed{\lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots \lambda_n y_n^2}$

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(10) Theorem:

i) The non-zero eigenvalues of A^TA and AA^T coincide

ii) Let X be the eigenvector corr. to non-zero eigenvalue λ

Then A^TX is the eigenvector of A^TA corr. to eigenvalue λ

Proof:

① Let λ be non-zero ~~and~~ ~~be~~ eigenvalue of AA^T

② $(AA^T)X = \lambda X$ ——ⓐ

multiply by A^T

$(A^T)(AA^T)X = \lambda (A^T)X$ ——ⓑ

$A^TA\ A^TX = \lambda\ A^TX$

③ $A^TX \neq 0$

If $A^TX = 0$, by ⓐ $\lambda X = 0$

$\lambda = 0$ or $X = 0$

not possible

④ from ②

$(A^TA)(A^TX) = \lambda (A^TX)$

Let $C = A^TA$

$V = A^TX$

then $CV = \lambda V$ ($\lambda \neq 0$, $V \neq 0$)

$\Rightarrow \therefore$ eigenvalue of $C = A^TA$ is λ corr. and eigenvector is $V = A^TX$.

(11) Singular Value Decomposition (SVD)

$A = U \Sigma V^T$

$A_{m \times n}$ $m < n$ wide

$U_{m \times m}$

$V^T_{n \times n}$

$\Sigma \rightarrow$ diagonal matrix

Proof:

① Let $A_{m \times n}$ be a wide matrix $m \leq n$

② $\therefore$ non-zero eigenvalues of AA^T and A^TA coincide as the matrices are symmetric and λ's are real.

③ Take $\lambda_1 \neq 0$,

$\therefore (AA^T)X = \lambda_1 X$ (non-zero eigenvalue of AA^T)

$(AA^T)U_1 = \lambda_1 U_1$ ——(i)

multiply by A^T

$A^T(AA^T)U_1 = \lambda_1 A^TU_1$

$(A^TA)A^TU_1 = \lambda_1 A^TU_1$

$(A^TA)V = \lambda_1 V$

V is eigenvector of A^TA for λ_1

④ Take $V_1 = \dfrac{A^TU_1}{\|A^TU_1\|}$ ——(ii)

⑤ Form $U = [U_1\ U_2\ U_3 \ldots U_n]_{m \times m}$

Form $V = [V_1\ V_2\ V_3 \ldots V_n]_{n \times n}$

⑥ $\|A^TU_1\|^2 = \langle A^TU_1, A^TU_1 \rangle$

$= (A^TU_1)^T A^TU_1$

$= U_1^T AA^T U_1$ ($\because U$ is unit vector.)

$= U_1^T \lambda_1 U_1$

$= \lambda_1 U_1^T U_1 = \lambda_1 \langle U_1, U_1 \rangle$

$= \lambda_1 \|U_1\|^2 = \lambda_1$

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$\therefore \|A^T u_1\| = \sqrt{\lambda_1}$ ——(iii)

⑦ Sub (iii) in (ii)

$$v_1 = \frac{A^T u_1}{\sqrt{\lambda_1}}$$

$$A v_1 = \frac{A A^T u_1}{\sqrt{\lambda_1}}$$

$$= \frac{\lambda_1 u_1}{\sqrt{\lambda_1}}$$

$$= \sqrt{\lambda_1}\, u_1$$

where $(\sigma = \sqrt{\lambda})$

$$A v_1 = \sigma_1 u_1$$

⑧ Similarly,

$$A v_2 = \sigma_2 u_2$$
$$A v_3 = \sigma_3 u_3$$
$$\vdots$$
$$A v_m = \sigma_m u_m$$
$$A v_{m+1} = 0 \ldots 0$$
$$\vdots$$
$$A v_n = 0$$

where $\sigma_i = \sqrt{\lambda_i}$

$\lambda_1 \geq \lambda_2 \geq \lambda_3 \ldots \geq 0$

⑨ vectorizing ⑧, we get.

$$A_{m\times n} \underset{n\times n}{[v_1\ v_2\ v_3 \cdots v_n]} = [u_1\ u_2 \ldots u_m] \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & 0 \\ & & \ddots & \\ 0 & & & \sigma_m \end{bmatrix}$$

⑩ $\therefore AV = U\Sigma$

$$A = U\Sigma V^{-1}$$

$$\boxed{A = U\Sigma V^T}$$

(V is orthogonal)

wide SVD of A.

⑪ Suppose B is a tall matrix $(m>n)$

Then B^T is wide $B^T_{n\times m}$

$\therefore$ SVD of $B^T = A$

$$AB^T = (U\Sigma V^T)^T$$

$$= V\Sigma$$

$$(A)^T = (B^T)^T = (U\Sigma V^T)^T$$

$$\boxed{B = V\Sigma^T U^T}$$ SVD of B Tall Matrix

⑫ Theorem:

Let $A_{m\times n}$ be a matrix of rank r

Then $A = U\Sigma V^T$ and,

(i) $(u_1, u_2, \ldots u_r)$ is a basis for $C(A) = R(A^T)$

(ii) $(u_{r+1}, u_{r+2} \cdots u_m)$ is a basis for $N(A^T)$

(iii) $(v_1, v_2, \ldots v_r)$ is a basis for $C(A^T) = R(A)$

(iv) $(v_{r+1}, v_{r+2} \ldots v_m)$ is a basis for $N(A)$

[i.e SVD computes basis for Four Fundamental Subspaces of Matrix]

Proof:

① $A = U\Sigma V^T$

$$A v_i = \sigma_i u_i$$

$$A^T(A v_i) = \sigma_i A^T u_i$$

$$\lambda_i v_i = \sigma_i A^T u_i$$

$$v_i = \frac{\sigma_i A^T u_i}{\lambda_i} \quad \text{for } i = 1,2,3,\ldots r$$

$$v_i = \left[\left(\frac{\sigma_i}{\lambda_i}\right) A^T u_1\right]$$ a linear combination of columns of A^T

$\Rightarrow v_i \in C(A^T) = R(A)$ $i = 1,2,3 \ldots r$

i.e proved (iii)

② Now, $AV_{r+1} = 0$, $AV_{r+2} = 0 \ldots AV_n = 0$

$\Rightarrow (v_{r+1}, v_{r+2}, \ldots v_n) \in N(A)$

(V is orthogonal)

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③ Now consider A^T

$$A = U \Sigma V^T$$

$$A^T = (U \Sigma V^T)^T$$

$$A^T = V \Sigma^T U^T$$

$$A^T U_i = \sigma_i V_i$$

$$A A^T U_i = \sigma_i A V_i$$

$$\lambda_i U_i = \sigma_i A V_i$$

$$U_i = \frac{\sigma_i}{\lambda_i} A V_i$$] a linear combination of columns of A

$$V_i \in C(A) = R(A^T) \quad \forall i = 1, 2, \ldots r$$

④ A^T [illegible] $= 0$

$$A^T U_{r+1} = 0, A^T U_{r+2} \ldots A^T U_m = 0$$

$$\therefore (U_{r+1}, U_{r+2}, \ldots U_m) \in N(A^T)$$

BASICS

① rank $(I_n) = n$

② rank (null matrix) $= 0$

③ rank $(A) =$ rank (A^T)

④ if $A \neq$ null, $r(A) \geq 1$

✱ SINGULAR det $= 0$

✱ NON SINGULAR det $\neq 0$

LINEAR TRANSFORMATION

✱ $T: V \to W$ if T:

addn $T(x+y) = T(x) + T(y)$ $\forall x,y \in V$

scalar multiplication $T(\alpha x) = \alpha T(x)$ $\forall x \in V$

$\Rightarrow T(\alpha X + \beta y) = \alpha T(x) + \beta T(y)$

EIGENVALUES
EIGENVECTORS

① $\boxed{AX = \lambda X}$

② $|A - \lambda I| = 0$ [Characteristic Eqn]

③ Eigenvalues $\Rightarrow$ values of λ after solving $|A - \lambda I| = 0$

④ Eigenvectors $\Rightarrow X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ for each corr. eigenvalue λ

⑤ for 2×2
$\Rightarrow \lambda^2 - tr(A)\lambda + |A| = 0$

⑥ for 3×3
$\Rightarrow \lambda^3 - tr(A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$

⑦ sum (eigenvalues) $= tr(A)$

⑧ prod (eigenvalues) $= \det(A)$

✱ if $A \to \lambda$, $A^{-1} \to \frac{1}{\lambda}$

Eigen Decomposition

$A = PDP^{-1}$ $\quad D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

$A^n = PD^nP^{-1}$ $\quad P = [x_1\ x_2 \cdots]$ ← eigenvectors

① Identity $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

② Scaling (abt x axis) $\begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix}$

③ Scaling (abt y-axis) $\begin{bmatrix} 1 & 0 \\ 0 & b \end{bmatrix}$

④ Shear (x-abt axis) $\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$

⑤ Shear (y-abt axis) $\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$

⑥ Rotation (a.c.w) $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} R_\theta$

⑦ Rotation (c.w) $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} R_{-\theta}$

⑧ Reflection (wrt x-axis) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

⑨ Reflection (wrt y-axis) $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

⑩ Reflection (abt a line L, angle θ) $\begin{bmatrix} \cos2\theta & \sin2\theta \\ \sin2\theta & -\cos2\theta \end{bmatrix}$

⑪ Projection (on x-axis) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

⑫ Projection (on y-axis) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

✱ ELEMENTARY DECOMPOSITION

eg: $\begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$

① Convert A into I

$A = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$

↓ ① $R_1 \to R_1/4$
② $R_2 \to R_2 - 2R_1$
③ $R_2 \to R_2 \times -2$
④ $R_1 \to R_1 - 3/4\, R_2$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

② Do individual row transformations on I to get E_i

$E_1 = R_1/4 \Rightarrow \begin{bmatrix} 1/4 & 0 \\ 0 & 1 \end{bmatrix}$

$E_2 = R_2 - 2R_1 \Rightarrow \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$

$E_3 = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$, $E_4 = \begin{bmatrix} 1 & -3/4 \\ 0 & 1 \end{bmatrix}$

③ Write in correct order

$E_4 \cdot E_3 \cdot E_2 \cdot E_1 A = I$

$\therefore A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} I$

④ Give geometric interpretation of $(E_i)^{-1}$ matrix.

✱ LU Decomposition

$LU = A$

↑ lower △, ↖ upper △

① Write eqn in terms of $Ax = b$

② Now write A as LU

$A = L \cdot U \cdot$
$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} [A]$

(i) Keep reducing U to its row echelon form to make upper △

(ii) for every transformation done on U, do -ve of it on L

③ $AX = b$

$LUX = b$

$\downarrow$

$\star \begin{bmatrix} p \\ q \end{bmatrix} = U \begin{bmatrix} x \\ y \end{bmatrix}$ — ✱

④ $L \begin{bmatrix} p \\ q \end{bmatrix} = [b]$ solve

⑤ Use p & q to solve ✱ eqn

(iii) ✱✱ do elementary opern not row operations.

PLU

P → permutation matrix

$P \cdot P^T = I = P^T \cdot P$

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \overset{R_2 \leftrightarrow R_1}{\longleftrightarrow} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} P_{12}$

✱ $\det(A) = \det(PLU)$

✱ $P_{12} B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$

left multiplication — rows exchanged

✱ $B P_{12} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix}$

right multiplication — columns exchanged

✱✱ used when first pivot to start LU is missing.

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ORTHOGONAL MAPPING IN 2D

Orthogonal if $A \cdot A^T = A^T \cdot A = I$
symmetric if $A^T = A$

* orthogonal maps in 2D are either reflection or rotation.

ROTATIONS IN 3D

Rot. abt x-axis (y-z plane) ~~PITCH~~ ROLL (α)

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rot abt y-axis (in z-x planes) PITCH (β)

$$\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

Rot abt z-axis (in x-y plane) YAW (γ)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

RANK-NULLITY THEOREM

① $\ker T = \{x \in V \mid T(x) = 0\}$ [Null Space]
② $\text{Im}\, T = \{T(x) \mid x \in V\}$ [Range space]

T is a Linear Transfn $T: V \to W$
→ $\ker(T) \subseteq V$ (dom)
→ $\text{Im}(T) \subseteq W$ (cod)

- $U \subseteq V$ if it is closed under addition and scalar multiplication.
- SUBSPACE should pass through the ORIGIN.

$\text{rank}(T) = \dim \text{Im}(T)$
$\text{nullity}(T) = \dim \text{Ker}(T)$
} Rank Nullity Theorem ⇒

$\text{rank}(T) + \text{nullity}(T) = \dim V$
$\dim(\text{Im}\, T) + \dim(\ker T) = \dim V$

① $T: P_n \to P_{n-1}$
$T(f) = f'$ is a L.T

$$m(T) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

DERIVATIVE ... n

② $T: P_n \to P_{n+1}$
$T(f) = \int f\, dx$ is a L.T

$$m(T) = \begin{bmatrix} c_1 & c_2 & c_3 \\ 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{bmatrix} \cdot \quad c_i \in \mathbb{R}$$

ANTIDERIVATIVE MATRIX.

FUNCTIONS + RNT

① Result 1:
$T: V \to W$ is a L.T
Then T is one-one iff $\ker T = \{0\}$.

② Result 2
$T: V \to W$ is a L.T
$\text{Im}(T) = \{T(x) \mid x \in V\} \subseteq W$
Then T is ONTO iff $\text{Im}(T) = W$

* ① T is one-one ⇔ $\ker(T) = \{0\}$
② T is onto ⇔ $\text{Im}(T) = W$
} $\dim V = \dim W$
③ T is a Linear Transformation $V \cong W$ (isomorphic to W)

* if $\ker T = \{0\}$ ⇒ dimensions as it is
* if $\ker(T) = x$ ⇒ dimensions reduced by x.